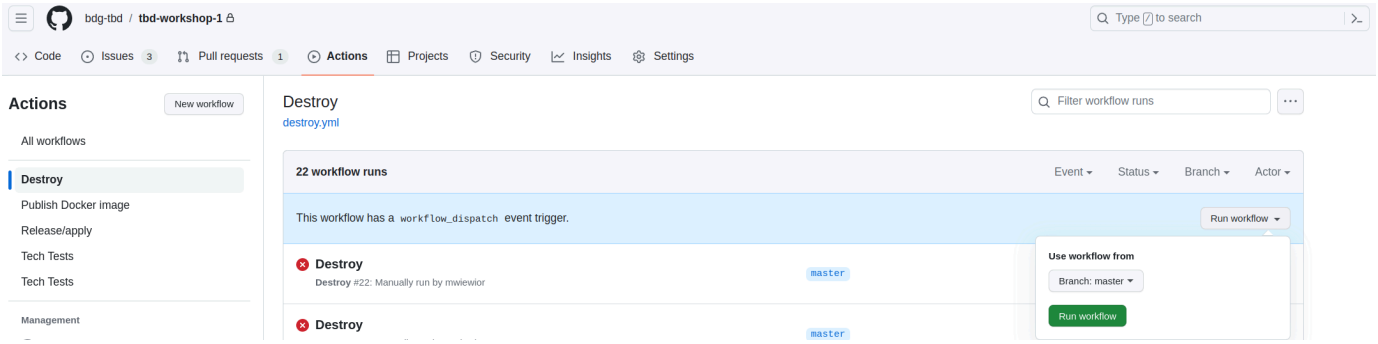
IMPORTANT ❗ ❗ ❗ Please remember to destroy all the resources after each work session. You can recreate infrastructure by creating new PR and merging it to master.



Phase 1 Exercise Overview

```
flowchart TD
    A[🔧 Step 0: Fork repository] --> B[🔧 Step 1: Environment variables\nexport TF_VAR_
    B --> C[🔧 Step 2: Bootstrap\nterraform init/apply\n→ GCP project + state bucket]
    C --> D[🔧 Step 3: Quota increase\nCPUS_ALL_REGIONS ≥ 24]
    D --> E[🔧 Step 4: CI/CD Bootstrap\nWorkload Identity Federation\n→ keyless auth GH→
    E --> F[🔧 Step 5: GitHub Secrets\nGCP_WORKLOAD_IDENTITY_*\nINFRACOST_API_KEY]
    F --> G[🔧 Step 6: pre-commit install]
    G --> H[🔧 Step 7: Push + PR + Merge\n→ release workflow\n→ terraform apply]

    H --> I{Infrastructure\nrunning on GCP}

    I --> J[📋 Task 3: Destroy\nGitHub Actions → workflow_dispatch]
    I --> K[📋 Task 4: New branch\nModify tasks-phase1.md\nPR → merge → new release]
    I --> L[📋 Task 5: Analyze Terraform\nterraform plan/graph\nDescribe selected module
    I --> M[📋 Task 6: YARN UI\ngcloud compute ssh\nIAP tunnel → port 8088]
    I --> N[📋 Task 7: Architecture diagram\nService accounts + buckets]
    I --> O[📋 Task 8: Infracost\nUsage profiles for\nartifact_registry + storage_bucket
    I --> P[📋 Task 9: Spark job fix\nAirflow UI → DAG → debug\nFix spark-job.py]
    I --> Q[📋 Task 10: BigQuery\nDataset + external table\non ORC files]
    I --> R[📋 Task 11: Spot instances\npreemptible_worker_config\nin Dataproc module]
    I --> S[📋 Task 12: Auto-destroy\nNew GH Actions workflow\nschedule + cleanup tag]

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```

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```

Legend

- 🔵 Blue — setup steps (one-time configuration)
- 🟠 Orange — manual steps (GCP Console / GitHub UI)
- 🟢 Green — infrastructure ready
- 🟣 Purple — tasks to complete and document in tasks-phase1.md

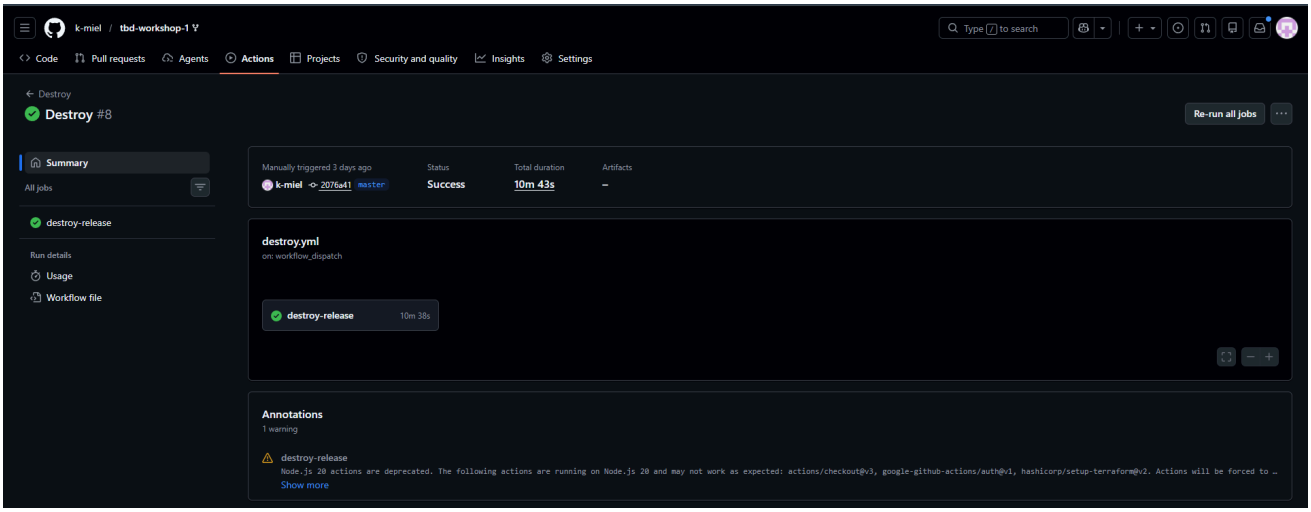
1. Authors:

   Group 2

   https://github.com/k-miel/tbd-workshop-1

2. Follow all steps in README.md.

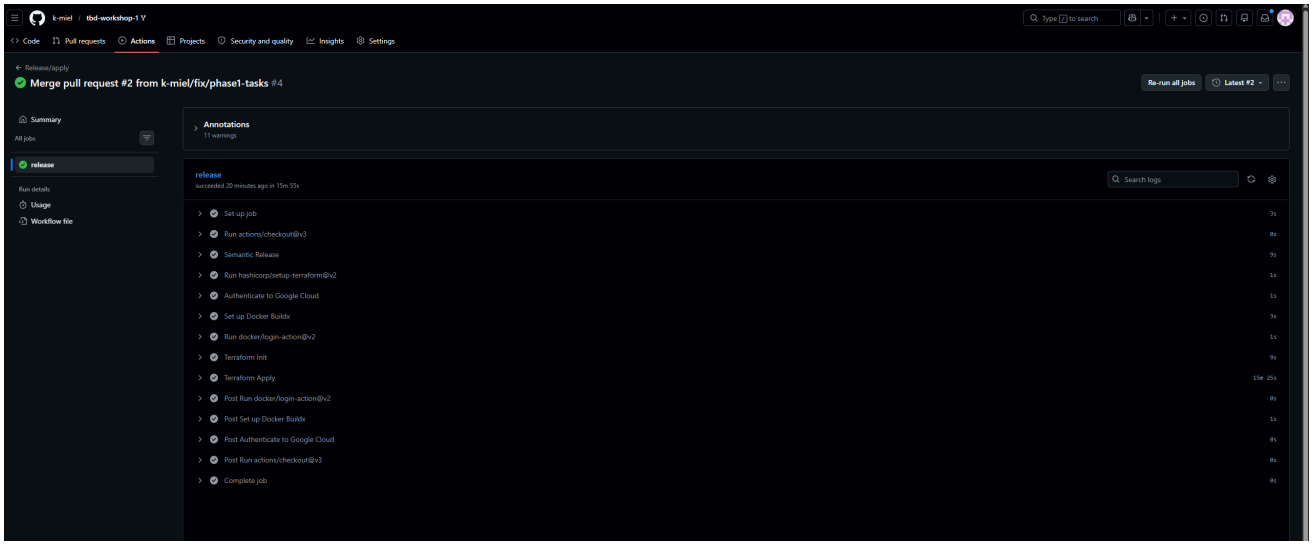
3. From available Github Actions select and run destroy on master branch.



4. Create new git branch and:

1. Modify tasks-phase1.md file.

2. Create PR from this branch to **YOUR** master and merge it to make new release.



5. Analyze terraform code. Play with terraform plan, terraform graph to investigate different modules.

Wybrany moduł: `modules/dataproc/` (main.tf)

Moduł `dataproc` tworzy kompletną infrastrukturę klastra Hadoop/Spark w GCP. Składa się z następujących zasobów:

Zasób Terraform	Opis
`google_project_service.dataproc`	Włącza API `dataproc.googleapis.com` w projekcie GCP
`google_service_account.dataproc_sa`	Tworzy dedykowany Service Account `{project_name}-dataproc-sa`
`google_project_iam_member.dataproc_worker`	Nadaje SA rolę `roles/dataproc.worker`
`google_project_iam_member.dataproc_bigquery_data_editor`	Nadaje SA rolę `roles/bigquery.dataEditor` (zapis danych)
`google_project_iam_member.dataproc_bigquery_user`	Nadaje SA rolę `roles/bigquery.user` (uruchamianie zapytań)
`google_storage_bucket.dataproc_staging`	Bucket GCS dla plików stagingowych zadań Spark

Zasób Terraform	Opis
`google_storage_bucket.dataproc_temp`	Bucket GCS dla plików tymczasowych zadań Spark
`google_storage_bucket_iam_member` (×2)	Przyznaje SA uprawnienia `roles/storage.objectAdmin` do obu bucketów
`google_dataproc_cluster.tbd-dataproc-cluster`	Główny klaster Dataproc (master + 2 worker + 2 preemptible)

Konfiguracja klastra `tbd-cluster`:

- Region: `europe-west1`, sieć: prywatna subnet `subnet-01` (10.10.10.0/24)
- `internal_ip_only = true` — klaster nie ma zewnętrznych IP, dostęp tylko przez IAP
- Komponenty opcjonalne: `JUPYTER`
- HTTP port access włączony (`enable_http_port_access = true`)
- Initialization action: instalacja przez pip: `pandas<2`, `mlflow`, `google-cloud-storage`, `jupyterlab`, `dbt-core`, `dbt-spark`
- Master: 1 × `e2-standard-2`, 100 GB pd-standard
- Workers: 2 × `e2-standard-2` + 2 preemptible, 100 GB pd-standard

Zależności zasobów (kolejność tworzenia wymuszona przez `depends_on` w klastrze):

- API i SA tworzone jako pierwsze (niezależnie od siebie)
- IAM roles i GCS Buckets tworzone równolegle (obydwa zależą od SA)
- Bucket IAM zależy od SA i GCS Buckets
- Cluster tworzony jako ostatni (depends_on: API, SA, IAM roles, Bucket IAM)

Output `terraform graph -type=plan -target=module.dataproc`:

```
digraph {
    compound = "true"
    newrank = "true"
    subgraph "root" {
        "[root] module.dataproc.google_dataproc_cluster.tbd-dataproc-cluster (expand
        "[root] module.dataproc.google_project_iam_member.dataproc_bigquery_data_edi
        "[root] module.dataproc.google_project_iam_member.dataproc_bigquery_user (ex
        "[root] module.dataproc.google_project_iam_member.dataproc_worker (expand)"
        "[root] module.dataproc.google_project_service.dataproc (expand)" [label = "
        "[root] module.dataproc.google_service_account.dataproc_sa (expand)" [label
        "[root] module.dataproc.google_storage_bucket.dataproc_staging (expand)" [la
        "[root] module.dataproc.google_storage_bucket.dataproc_temp (expand)" [label
        "[root] module.dataproc.google_storage_bucket_iam_member.staging_bucket_iam
        "[root] module.dataproc.google_storage_bucket_iam_member.temp_bucket_iam (ex
```

```
        "[root] module.dataproc.google_dataproc_cluster.tbd-dataproc-cluster (expand
        "[root] module.dataproc.google_dataproc_cluster.tbd-dataproc-cluster (expand
        "[root] module.dataproc.google_dataproc_cluster.tbd-dataproc-cluster (expand
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        "[root] module.dataproc.google_project_iam_member.dataproc_bigquery_data_edi
        "[root] module.dataproc.google_project_iam_member.dataproc_bigquery_user (ex
        "[root] module.dataproc.google_project_iam_member.dataproc_worker (expand)"
        "[root] module.dataproc.google_storage_bucket_iam_member.staging_bucket_iam
        "[root] module.dataproc.google_storage_bucket_iam_member.staging_bucket_iam
        "[root] module.dataproc.google_storage_bucket_iam_member.temp_bucket_iam (ex
        "[root] module.dataproc.google_storage_bucket_iam_member.temp_bucket_iam (ex
    }
}
```

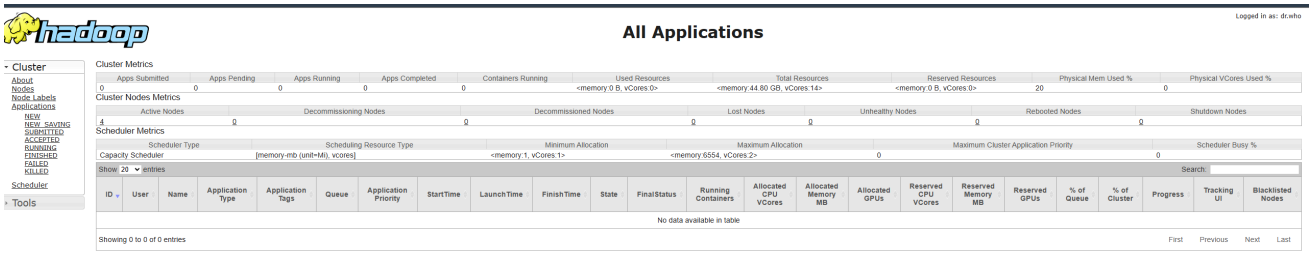
6. Reach YARN UI

Komenda tunelu IAP:

```
gcloud compute ssh tbd-cluster-m \
  --project=tbd-2026l-321362 \
  --zone=europe-west1-b \
  --tunnel-through-iap \
  -- -L 8088:localhost:8088 -N
```

Port: 8088 (YARN ResourceManager UI)

Opis: Klaster Dataproc ma `internal_ip_only = true`, więc nie posiada zewnętrznego IP. Połączenie SSH przez `--tunnel-through-iap` kieruje ruch przez Google Identity-Aware Proxy (firewall `fw-allow-ingress-iap` przepuszcza TCP:22 z zakresu `35.235.240.0/20`). Flaga `-L 8088:localhost:8088` tworzy lokalny port-forward — po uruchomieniu YARN UI jest dostępny pod adresem **http://localhost:8088**.

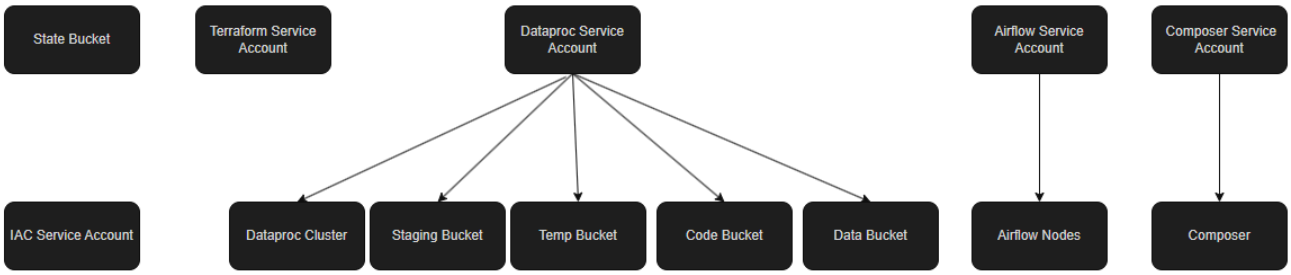


Hint: the Dataproc cluster has `internal_ip_only = true`, so you need to use an IAP tunnel. See: `gcloud compute ssh` with `-- -L <local_port>:localhost:<remote_port>` and `--tunnel-through-iap` flag. YARN ResourceManager UI runs on port **8088**.

7. Draw an architecture diagram (e.g. in draw.io) that includes:

    1. Description of the components of service accounts
    2. List of buckets for disposal

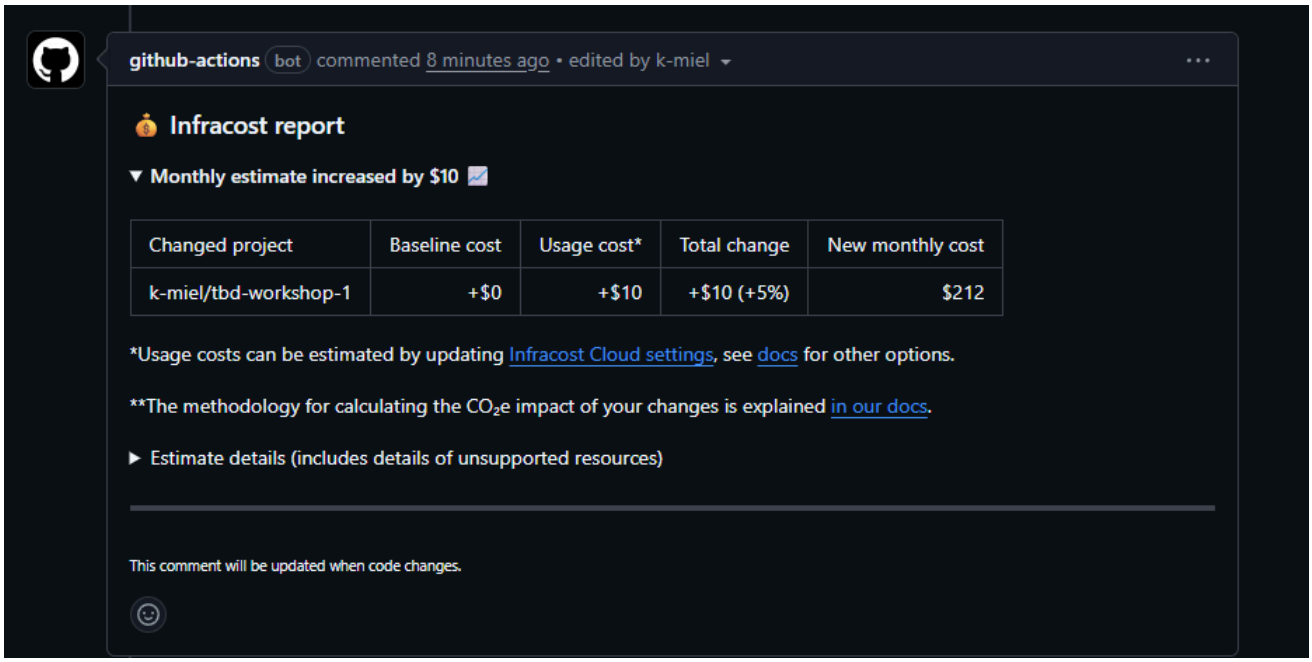
    Diagram architektury:



8. Create a new PR and add costs by entering the expected consumption into Infracost For all the resources of type: `google_artifact_registry_repository`, `google_storage_bucket` create a sample usage profiles and add it to the Infracost task in CI/CD pipeline. Usage file [example]

```
 version: 0.1
resource_usage:
  module.gcr.google_artifact_registry_repository.registry:
    storage_gb: 150
    monthly_data_transfer_gb: 10
  module.data-pipelines.google_storage_bucket.tbd-data-bucket:
    storage_gb: 500
    monthly_data_retrieval_gb: 100
    monthly_egress_data_transfer_gb: 100
  module.data-pipelines.google_storage_bucket.tbd-code-bucket:
    storage_gb: 10
    monthly_data_retrieval_gb: 5
    monthly_egress_data_transfer_gb: 5
  module.dataproc.google_storage_bucket.dataproc_staging:
    storage_gb: 50
    monthly_data_retrieval_gb: 20
    monthly_egress_data_transfer_gb: 10
  module.dataproc.google_storage_bucket.dataproc_temp:
```

```
    storage_gb: 50
    monthly_data_retrieval_gb: 20
    monthly_egress_data_transfer_gb: 10
```



9. Find and correct the error in spark-job.py

   After `terraform apply` completes, connect to the Airflow cluster:

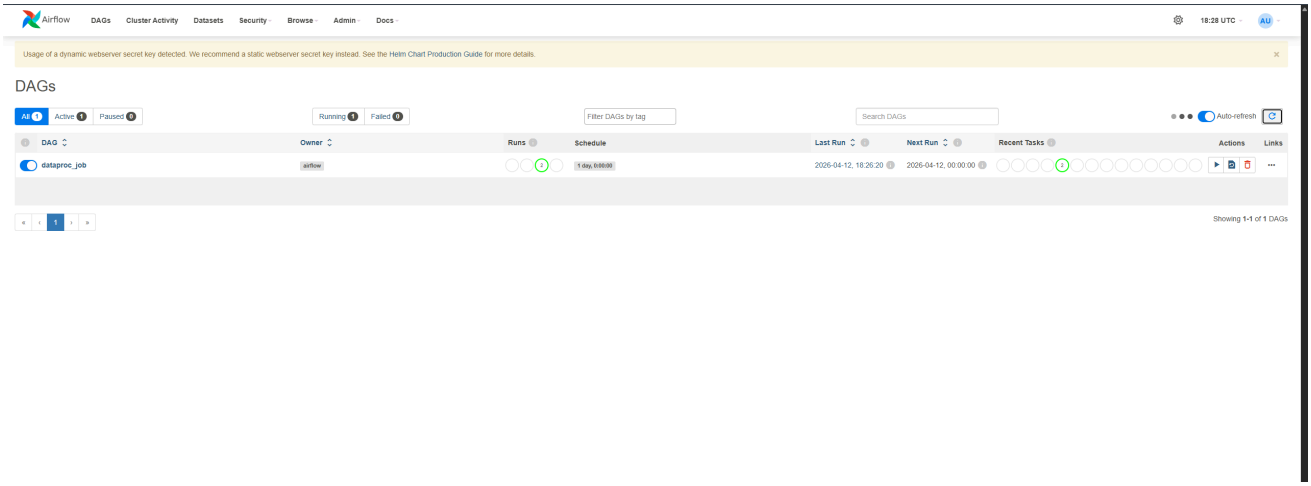
```
gcloud container clusters get-credentials airflow-cluster --zone europe-west1-b --p
```

   Then check the external IP (AIRFLOW_EXTERNAL_IP) of the webserver service: kubectl get svc -n airflow airflow-webserver

   > Note: If EXTERNAL-IP shows , wait a moment and retry — LoadBalancer IP allocation may take 1-2 minutes.

   DAG files are synced automatically from your GitHub repo via git-sync sidecar. Airflow variables and the `google_cloud_default` GCP connection are also configured by Terraform.

   a) In the Airflow UI (http://AIRFLOW_EXTERNAL_IP:8080, login: admin/admin), find the `dataproc_job` DAG, unpause it and trigger it manually.



b) The DAG will fail. Examine the task logs in the Airflow UI to find the root cause.

```
 google.api_core.exceptions.NotFound: 404 GET https://storage.googleapis.com/storage
The specified bucket does not exist.
```

Błąd polegał na nieprawidłowym identyfikatorze projektu zakodowanym na stałe w skrypcie `spark-job.py` — ścieżka wyjściowa wskazywała na bucket `gs://tbd-2026l-9010-data/`, który nie istnieje. Prawidłowa nazwa bucketu to `gs://tbd-2026l-321362-data/`. Błąd znaleziono w logach taska `submit_dataproc_job` w Airflow UI (zakładka *Log* → sekcja *Exception*).

c) Fix the error in `modules/data-pipeline/resources/spark-job.py` and re-upload the file to GCS:

```
 gsutil cp modules/data-pipeline/resources/spark-job.py gs://PROJECT_NAME-code/spark
```

Then trigger the DAG again from the Airflow UI.

[modules/data-pipeline/resources/spark-job.py]

d) Verify the DAG completes successfully and check that ORC files were written to the data bucket:

```
 gs://tbd-2026l-321362-data/data/shakespeare/
gs://tbd-2026l-321362-data/data/shakespeare/_SUCCESS
gs://tbd-2026l-321362-data/data/shakespeare/part-00000-4ec4b2b9-143b-4e12-b8bc-cadcc
gs://tbd-2026l-321362-data/data/shakespeare/part-00000-afad560a-fa79-4481-a7b7-4a61d
gs://tbd-2026l-321362-data/data/shakespeare/part-00001-4ec4b2b9-143b-4e12-b8bc-cadcc
gs://tbd-2026l-321362-data/data/shakespeare/part-00001-afad560a-fa79-4481-a7b7-4a61d
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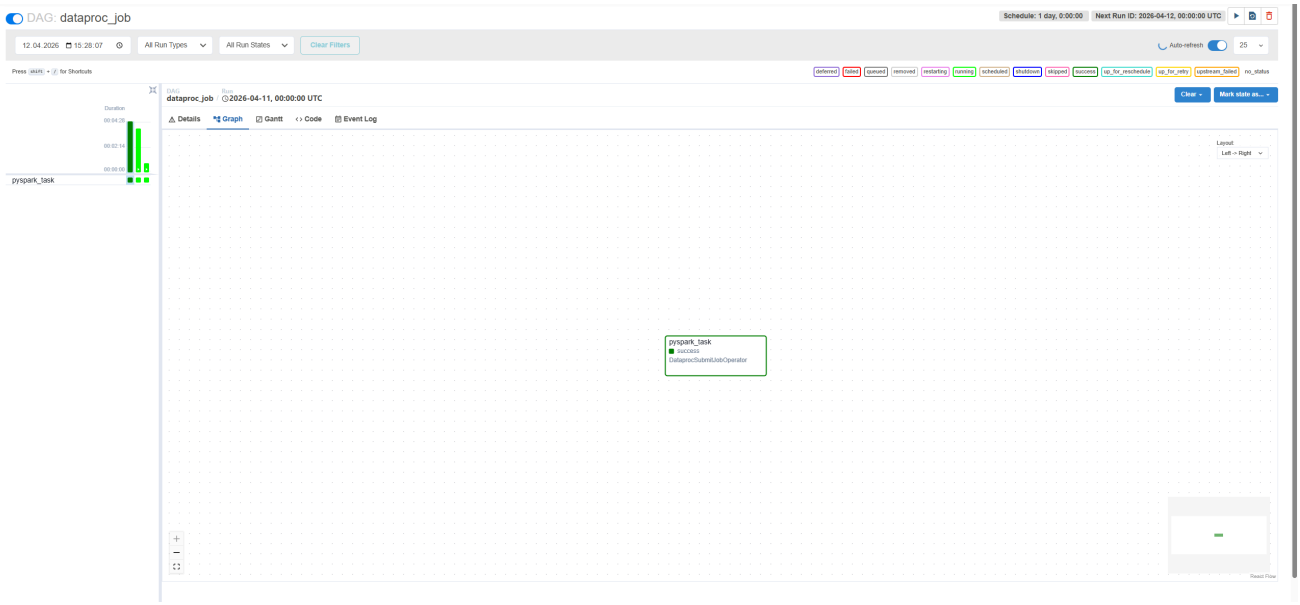
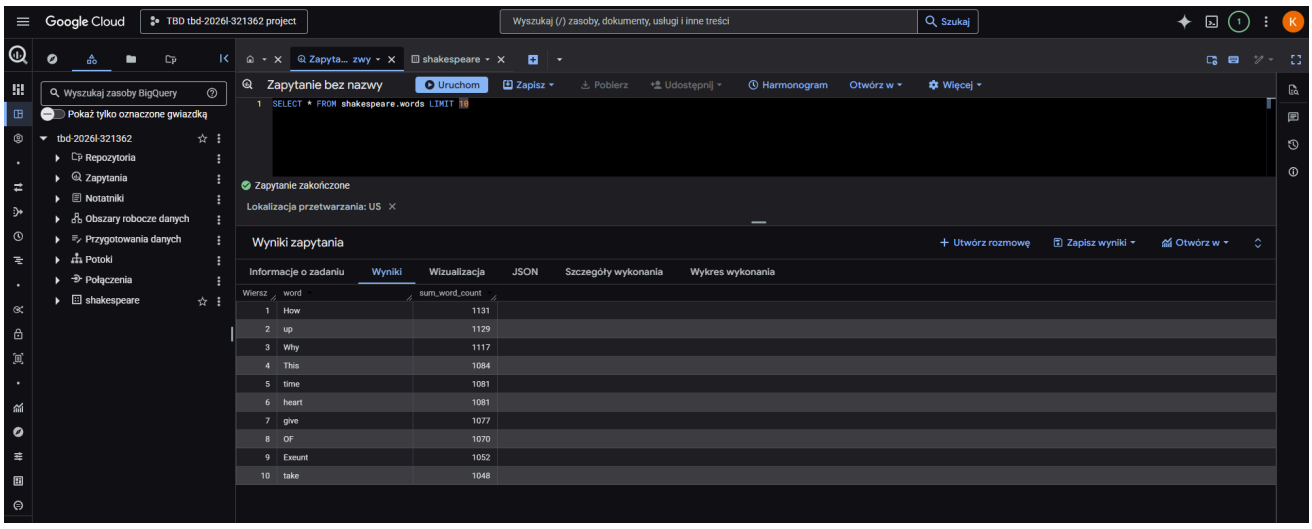
gs://tbd-2026l-321362-data/data/shakespeare/part-00002-afad560a-fa79-4481-a7b7-4a61d
gs://tbd-2026l-321362-data/data/shakespeare/part-00003-4ec4b2b9-143b-4e12-b8bc-cadcc
```



10. Create a BigQuery dataset and an external table using SQL

```
 CREATE SCHEMA IF NOT EXISTS shakespeare;
CREATE OR REPLACE EXTERNAL TABLE shakespeare.words
OPTIONS (
  format = 'ORC',
  uris = ['gs://tbd-2026l-321362-data/data/shakespeare/*.orc']
);
```



why does ORC not require a table schema? ORC (Optimized Row Columnar) jest formatem "self-describing". Oznacza to, że pliki ORC przechowują metadane dotyczące schematu (nazwy i typy kolumn) bezpośrednio wewnątrz swojej struktury (zwykle w stopce pliku - footer). Dzięki temu systemy takie jak BigQuery potrafią automatycznie wydobyć strukturę tabeli bez konieczności jej ręcznego definiowania w zapytaniu DDL.

11. Add support for preemptible/spot instances in a Dataproc cluster

Link do zmodyfikowanego pliku: [modules/dataproc/main.tf]

Wstawiony kod Terraform (fragment `cluster_config` w zasobie `google_dataproc_cluster`):

```
 preemptible_worker_config {
  num_instances = 2
}
```

12. Triggered Terraform Destroy on Schedule or After PR Merge. Goal: make sure we never forget to clean up resources and burn money.

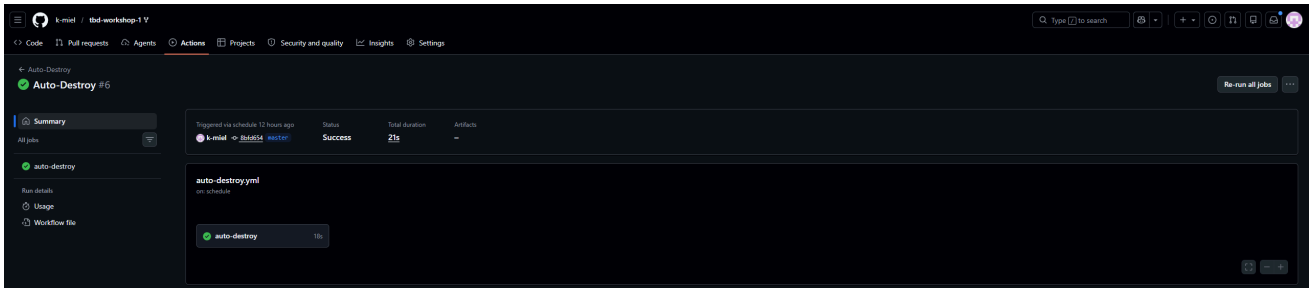
```
 name: Auto-Destroy
on:
  schedule:
    - cron: '0 20 * * *'
  pull_request:
    branches: [ master ]
    types: [ closed ]

permissions: read-all

jobs:
  auto-destroy:
    if: |
      github.event_name == 'schedule' ||
      (github.event.pull_request.merged == true && contains(github.event.pull_reques
    runs-on: ubuntu-latest
    permissions:
      contents: read
      id-token: write

    steps:
      - uses: 'actions/checkout@v3'
      - uses: hashicorp/setup-terraform@v2
        with:
          terraform_version: 1.11.0
      - id: 'auth'
        name: 'Authenticate to Google Cloud'
        uses: 'google-github-actions/auth@v1'
        with:
          workload_identity_provider: ${{ secrets.GCP_WORKLOAD_IDENTITY_PROVIDER_NAM
          service_account: ${{ secrets.GCP_WORKLOAD_IDENTITY_SA_EMAIL }}
      - name: Terraform Init
```

```
        run: terraform init -backend-config=env/backend.tfvars
      - name: Terraform Destroy
        run: terraform destroy -auto-approve -var-file env/project.tfvars
```



write one sentence why scheduling cleanup helps in this workshop Automatyczne niszczenie zasobów zgodnie z harmonogramem zapobiega niepotrzebnemu zużywaniu środków na koncie GCP w przypadku zapomnienia o ręcznym wywołaniu komendy destroy po zakończeniu pracy.